

Exercise 6: Cursors

Scenario 1: Generate monthly statements for all customers.

Question: Write a PL/SQL block using an explicit cursor

GenerateMonthlyStatements that retrieves all transactions for the current month and prints a statement for each customer.

Program

DECLARE

CURSOR GenerateMonthlyStatements IS

SELECT c.Name,

t.TransactionDate,

t.Amount,

t.TransactionType

FROM Customers c

JOIN Accounts a

ON c.CustomerID = a.CustomerID

JOIN Transactions t

ON a.AccountID = t.AccountID

WHERE TO_CHAR(t.TransactionDate,'MMYYYY') =

TO_CHAR(SYSDATE,'MMYYYY');

v_name Customers.Name%TYPE;

v_date Transactions.TransactionDate%TYPE;

v_amount Transactions.Amount%TYPE;

v_type Transactions.TransactionType%TYPE;

BEGIN

OPEN GenerateMonthlyStatements;

LOOP

FETCH GenerateMonthlyStatements

INTO v_name, v_date, v_amount, v_type;

EXIT WHEN GenerateMonthlyStatements%NOTFOUND;

DBMS_OUTPUT.PUT_LINE(

'Customer: ' || v_name ||

' | Date: ' || TO_CHAR(v_date,'DD-MON-YYYY') ||

' | Type: ' || v_type ||

' | Amount: ' || v_amount

);

END LOOP;

CLOSE GenerateMonthlyStatements;

END;

Scenario 2: Apply annual fee to all accounts.

Question: Write a PL/SQL block using an explicit cursor **ApplyAnnualFee** that deducts an annual maintenance fee from the balance of all accounts.

Program

DECLARE

CURSOR ApplyAnnualFee IS

SELECT AccountID

FROM Accounts;

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v_accountID Accounts.AccountID%TYPE;

BEGIN

OPEN ApplyAnnualFee;

LOOP

    FETCH ApplyAnnualFee

    INTO v_accountID;

    EXIT WHEN ApplyAnnualFee%NOTFOUND;

    UPDATE Accounts

    SET Balance = Balance - 500

    WHERE AccountID = v_accountID;

END LOOP;

CLOSE ApplyAnnualFee;

COMMIT;

DBMS_OUTPUT.PUT_LINE('Annual maintenance fee applied successfully!');

EXCEPTION

    WHEN OTHERS THEN

        ROLLBACK;

        DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);

END;
```

Scenario 3: Update the interest rate for all loans based on a new policy.

Question: Write a PL/SQL block using an explicit cursor **UpdateLoanInterestRates** that fetches all loans and updates their interest rates based on the new policy.

Program

DECLARE

CURSOR UpdateLoanInterestRates IS

SELECT LoanID,

InterestRate

FROM Loans;

v_loanID Loans.LoanID%TYPE;

v_interest Loans.InterestRate%TYPE;

BEGIN

OPEN UpdateLoanInterestRates;

LOOP

FETCH UpdateLoanInterestRates

INTO v_loanID, v_interest;

EXIT WHEN UpdateLoanInterestRates%NOTFOUND;

UPDATE Loans

SET InterestRate = v_interest + 0.5

WHERE LoanID = v_loanID;

END LOOP;

CLOSE UpdateLoanInterestRates;

COMMIT;

DBMS_OUTPUT.PUT_LINE('Loan interest rates updated successfully.');

EXCEPTION

WHEN OTHERS THEN

ROLLBACK;

DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);

END;